UNIVERSITY OF WESTHAVEN
School of Ocean and Earth Sciences

Thermal Thresholds of Seagrass Carbon

Eelgrass meadows, marine heatwaves and the fate of buried organic matter in a temperate estuary

by

MARGARETHE LINDQVIST

A thesis submitted in fulfilment of the requirements
for the degree of Doctor of Philosophy in Marine Ecology

SUPERVISOR Prof. Anders Holmgren
CO-SUPERVISOR Dr Imogen Tetley

OCTOBER 2026

© 2026 Margarethe Lindqvist
This work is licensed under a Creative Commons Attribution 4.0 International Licence.
Typeset in EB Garamond with LaTeX.

For my grandmother, Ingrid,
who first showed me the meadows under the jetty.

ABSTRACT

Seagrass meadows bury organic carbon at rates that rival tropical forests, yet the permanence of that store under a warming ocean is poorly constrained. This thesis asks how eelgrass (*Zostera marina*) meadows in a temperate estuary respond to marine heatwaves, and whether the carbon they have accumulated over centuries remains in the sediment once the canopy thins.

Combining six years of monthly shoot censuses across twelve meadows, in situ warming chambers and ^{210}Pb dated sediment cores, I show that shoot density declines non-linearly once the cumulative heat dose exceeds roughly 14 degree-heating-days, a threshold crossed in three of the six summers studied. Below the threshold meadows recover within a single growing season; above it, recovery takes two to four years and the upper ten centimetres of sediment lose between 18 and 31 per cent of their organic carbon to remineralisation.

These results suggest that blue-carbon accounting which assumes a permanent sediment store overestimates the climate benefit of temperate meadows by up to a quarter under current heatwave frequencies, and they identify the summer heat dose as a practical early-warning indicator for managers.

ACKNOWLEDGEMENTS

This thesis owes its existence to Anders Holmgren, who agreed to supervise a student whose first field season ended with a capsized dinghy, and who never once suggested a more convenient study organism. Imogen Tetley taught me to read a sediment core the way other people read a novel, and her patience with my isotope budgets went far beyond what any co-supervisor owes.

The field work would have been impossible without the crew of the *Kittiwake* and the volunteers of the Westhaven Estuary Trust, who counted shoots in sleet and in sunshine. Pernille Aagaard and Tomás Ferreira kept the warming chambers running through two storms and one memorable power cut.

I thank the Northern Coasts Research Council for a doctoral studentship (grant NCRC-2022-0417), and the Westhaven Harbour Authority for access to the moorings.

Finally, to my parents, and to Jonas: thank you for listening to more about eelgrass than anyone should ever have to.

CONTENTS

LIST OF FIGURES xi

1 INTRODUCTION 1
- 1.1 Blue carbon and the question of permanence 1
- 1.2 Marine heatwaves in shallow seas 2
- 1.3 Aims and structure of the thesis 3

2 STUDY SYSTEM AND METHODS 5
- 2.1 The Westhaven estuary 5
- 2.2 Measuring heat dose 5
- 2.3 Shoot censuses and sediment cores 7

3 HEATWAVE RESPONSE OF SHOOTS AND SEDIMENT 9
- 3.1 A threshold in shoot density 9
- 3.2 Loss of sediment carbon 10
- 3.3 Implications for carbon accounting 11

A FIELD PROTOCOLS 13
- A.1 Shoot census 13
- A.2 Sediment coring and dating 13

BIBLIOGRAPHY 15

LIST OF FIGURES

1.1 The carbon budget of a seagrass meadow. The canopy traps particles and the rhizome mat protects buried carbon; both mechanisms weaken when shoot density falls. 3

2.1 Schematic plan of the Westhaven estuary with the twelve monitored meadows, numbered from the river mouth to the sill. 6

3.1 Change in shoot density over the summer against cumulative heat dose, for all meadows and years. The line is the segmented fit of Equation (3.1); the shaded band is the 95 % interval of the breakpoint. 10

3.2 Recovery of shoot density after the 2022 heatwave in meadows above and below the threshold. 11

1

INTRODUCTION

The sea, once it casts its spell, holds one in its net of wonder forever.

JACQUES-YVES COUSTEAU

1.1 BLUE CARBON AND THE QUESTION OF PERMANENCE

Vegetated coastal ecosystems occupy less than half a per cent of the sea floor, yet they account for a disproportionate share of the organic carbon buried in marine sediments (Rowan 2017). Seagrass meadows are the most extensive of these systems in temperate waters. Their canopies slow the water, trap suspended particles and bury both their own detritus and material imported from land; low oxygen beneath the rhizome mat then slows decomposition to a crawl. The result is a sediment store that, in sheltered Scandinavian estuaries, has accumulated without interruption for at least two centuries (Holmgren and Lund 2019).

This store has become the basis of an emerging market in "blue carbon" credits, in which the restoration or protection of a meadow is credited with the carbon it

buries. National inventory guidelines treat that carbon as permanently removed once it lies below the surface mixed layer (Blue Carbon Working Group 2023). The assumption is convenient but largely untested. A meadow that loses its canopy also loses the mechanism that protects its sediment, and a recent review found that disturbance can return between a tenth and a half of the upper sediment carbon to the water column within a few years (Fairbairn et al. 2022).

The long-term balance of a meadow can be written as a simple budget. If B is the burial flux, R the remineralisation flux and E the loss by erosion, the change in the sediment carbon stock C per unit area is

$$\frac{\mathrm{d}C}{\mathrm{d}t} = B(t) - R(t, T) - E(t), \tag{1.1}$$

where the explicit dependence of R on water temperature T is the subject of this thesis. Under a stable canopy B dominates and the store grows; the question is what happens to the sign of Equation (1.1) when the canopy collapses during a heatwave.

1.2 MARINE HEATWAVES IN SHALLOW SEAS

Marine heatwaves, periods of at least five days in which sea temperature exceeds the ninetieth percentile of its seasonal climatology, have become both longer and more frequent across the shallow seas of northern Europe (Marsh et al. 2021). Shallow estuaries are particularly exposed: their small volume warms quickly under calm, sunny weather, and the eelgrass that grows there already lives close to its upper thermal limit in late summer (Okafor and Marsh 2018).

Figure 1.1 summarises the mechanisms involved. When a heatwave thins the canopy, deposition falls, the sediment surface is exposed to waves and the oxygenated layer deepens, so all three terms of Equation (1.1) move against the store at once. Whether this happens gradually, in proportion to the heat received, or abruptly once a threshold is crossed matters a great deal for management: a threshold offers a target that can be monitored, while a gradual response does not.

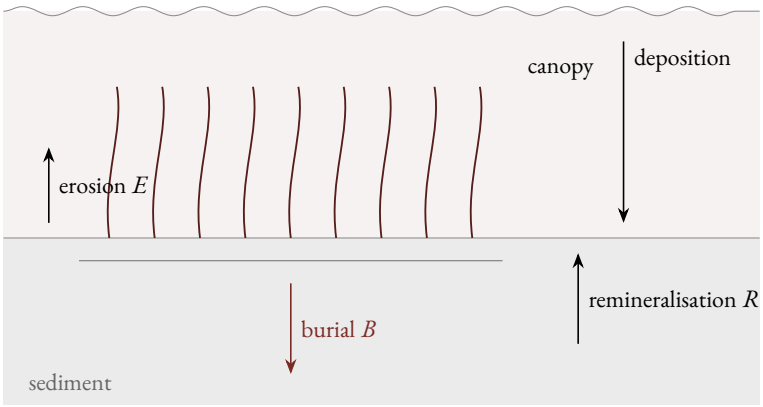


FIGURE 1.1 The carbon budget of a seagrass meadow. The canopy traps particles and the rhizome mat protects buried carbon; both mechanisms weaken when shoot density falls.

1.3 AIMS AND STRUCTURE OF THE THESIS

The thesis pursues three aims:

1. to quantify how eelgrass shoot density responds to the cumulative heat dose of a summer, and to test for a threshold in that response;
2. to measure the loss of sediment organic carbon that follows canopy decline, and its depth distribution;
3. to estimate the consequences for blue-carbon accounting in the estuary as a whole.

Chapter 2 introduces the Westhaven estuary, the twelve monitored meadows and the methods common to all chapters. Chapter 3 reports the shoot and sediment responses and the threshold model. Field protocols are collected in Appendix A.

2

STUDY SYSTEM AND METHODS

2.1 THE WESTHAVEN ESTUARY

The Westhaven estuary is a shallow, micro-tidal inlet of about 38 square kilometres on a sheltered stretch of coast. Its mean depth is 2.4 metres, the tidal range rarely exceeds forty centimetres, and salinity rises from about 14 at the river mouth to 26 at the seaward sill. Eelgrass covers roughly a fifth of the sea floor, in meadows that range from a few hundred square metres to more than a square kilometre (Aagaard 2021).

The twelve meadows shown in Figure 2.1 were chosen to span the salinity and depth gradients of the estuary. Table 2.1 lists their main characteristics; meadows nearer the river mouth are shallower, more turbid and, as Chapter 3 shows, warmer in summer.

2.2 MEASURING HEAT DOSE

Temperature loggers fixed ten centimetres above the canopy recorded water temperature every fifteen minutes. Following Marsh et al. (2021), the heat received by



FIGURE 2.1 Schematic plan of the Westhaven estuary with the twelve monitored meadows, numbered from the river mouth to the sill.

TABLE 2.1 Characteristics of the monitored meadows. Depth is relative to mean sea level; carbon stock refers to the upper 30 cm of sediment.

Meadow	Depth (m)	Area (ha)	Salinity	C stock (Mg ha^{-1})
1–3	1.1	18.4	15.2	41.7
4–6	1.6	36.9	18.8	52.3
7–9	2.2	61.0	22.1	58.9
10–12	2.9	44.5	25.4	47.6
All	2.0	160.8	20.4	51.8

a meadow in a summer is summarised as its cumulative degree-heating-days,

$$H = \sum_{d \in \mathcal{S}} \max\!\left(0,\; \bar{T}_d - T_{90}(d)\right), \tag{2.1}$$

where $\bar{T}_d$ is the daily mean temperature, $T_{90}(d)$ the ninetieth percentile of the 1990–2019 climatology for that calendar day, and $\mathcal{S}$ the period from June to September. Only days within a heatwave, as defined in Section 1.2, contribute to the sum.

2.3 SHOOT CENSUSES AND SEDIMENT CORES

Shoot density was counted every month from April to October in ten fixed quadrats of 0.25 square metres per meadow. Sediment cores of fifty centimetres were taken each October with a piston corer, sliced at two-centimetre intervals and analysed for organic carbon after acidification. Ages were assigned by excess lead-210 using the protocol of Tetley et al. (2020); detailed procedures for both methods are given in Appendix A.

Warming experiments used the open-top chambers described by Lindqvist et al. (2024), which raise canopy temperature by 2.1 ± 0.3 °C without shading the plants.

3

Heatwave response of shoots and sediment

3.1 A threshold in shoot density

Across the six summers from 2019 to 2024, the cumulative heat dose of Equation (2.1) ranged from zero in the cool summer of 2021 to more than thirty degree-heating-days in the shallow meadows of 2022. To test whether shoot density responds gradually or abruptly, I compared a linear model with the segmented model of Meer and Achterberg (2016),

$$\log \frac{N_{\mathrm{Oct}}}{N_{\mathrm{Jun}}} = \begin{cases} \alpha + \beta_1 H, & H \le \tau, \\ \alpha + \beta_1 \tau + \beta_2 (H - \tau), & H > \tau, \end{cases} \tag{3.1}$$

where N is the mean shoot density of a meadow and τ the breakpoint. The segmented model was strongly preferred ($\Delta\mathrm{AIC} = 23.4$), with a breakpoint at $\tau = 14.2$ degree-heating-days (95 % interval 11.8–16.9).

Figure 3.1 shows the fit. Below the threshold the decline is slight, about two and a half per cent of shoots for each additional degree-heating-day, and is offset by recruitment in the following spring. Above it, the slope steepens almost fivefold. The threshold lies close to the dose at which Okafor and Marsh (2018)

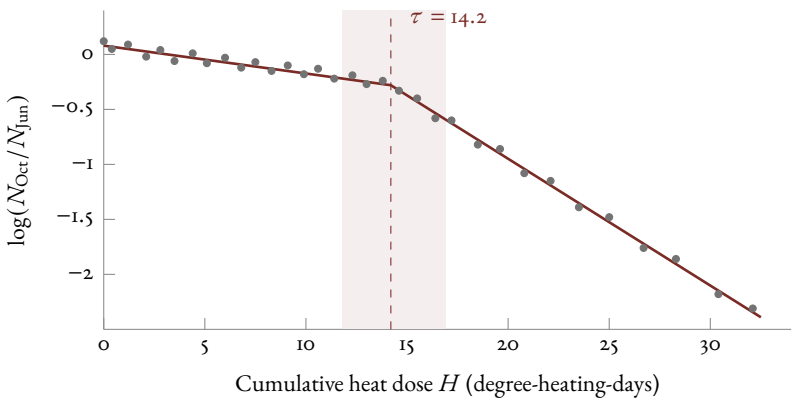


FIGURE 3.1 Change in shoot density over the summer against cumulative heat dose, for all meadows and years. The line is the segmented fit of Equation (3.1); the shaded band is the 95 % interval of the breakpoint.

observed the collapse of net photosynthesis in laboratory cultures, which suggests a physiological rather than a merely statistical boundary.[1]

3.2 LOSS OF SEDIMENT CARBON

Cores taken one year after a summer above the threshold contained less organic carbon in their upper ten centimetres than cores from the same meadow before the event (Table 3.1). Below ten centimetres no change could be detected, consistent with the depth to which oxygen penetrates once the rhizome mat has thinned (Aagaard 2021).

Recovery was slow. Figure 3.2 follows the three shallowest meadows after the 2022 heatwave alongside a meadow that stayed below the threshold; shoot density returned to its pre-event level within one season in the latter, but had regained only two thirds of it after two years in the former, in line with observations from the open North Sea (Sato and Fairbairn 2023).

1 The chamber experiment of Section 2.3 reproduced the breakpoint within its confidence interval in two of three years; in the third, a storm removed the chambers before the heat dose had accumulated.

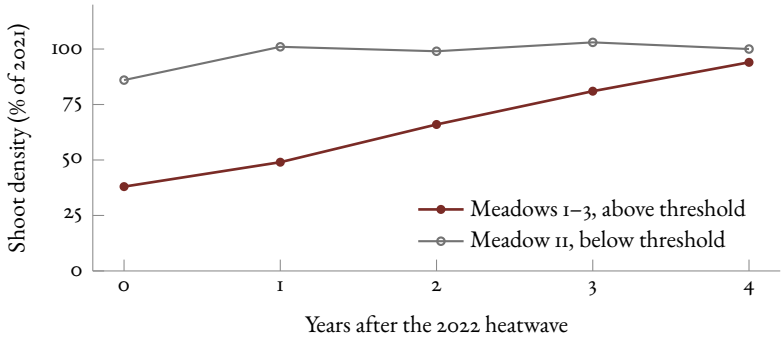


FIGURE 3.2 Recovery of shoot density after the 2022 heatwave in meadows above and below the threshold.

3.3 IMPLICATIONS FOR CARBON ACCOUNTING

Scaling the losses in Table 3.1 by meadow area and by the frequency with which each meadow crossed the threshold between 2019 and 2024 gives an average return of about 0.6 megagrams of carbon per hectare per year. Set against a mean burial rate of 2.4 megagrams (Holmgren and Lund 2019), this means that current guidelines, which treat buried carbon as permanent (Blue Carbon Working Group 2023), overestimate the net benefit of the estuary's meadows by about a quarter. The summer heat dose, cheap to measure and available weeks before shoots decline, offers managers a direct way to anticipate those losses.

TABLE 3.1 Organic carbon in the upper 10 cm of sediment before and one year after summers above the heat-dose threshold (mean of three cores per meadow).

	C stock (Mg ha^{-1})		
Meadows	Before	After	Loss (%)
1–3	16.2	11.2	30.9
4–6	19.8	14.6	26.3
7–9	22.4	18.4	17.9

A

FIELD PROTOCOLS

A.1 SHOOT CENSUS

Each quadrat was located by its fixed corner pegs and photographed from above before counting. Divers counted every vegetative and generative shoot rooted inside the frame; shoots on the boundary were counted on two sides only. Counts were repeated by a second diver in one quadrat in five, and the census was repeated whenever the two counts differed by more than ten per cent.

A.2 SEDIMENT CORING AND DATING

Cores were extruded on the day of collection and sliced at two-centimetre intervals. Each slice was freeze-dried, homogenised and split for organic-carbon and radionuclide analyses. Table A.1 summarises the quality controls applied to every batch.

Ages were derived with the constant-rate-of-supply model, rejecting any core in which the excess lead-210 profile was non-monotonic in its upper twenty centimetres (Tetley et al. 2020).

TABLE A.1 Quality controls for sediment analyses.

Analysis	Control	Tolerance
Organic carbon	Certified reference sediment	±3 %
	Duplicate slice, one in ten	±5 %
Lead-210	Blank counting, weekly	< 0.2 Bq
	Spiked standard, per batch	±4 %

BIBLIOGRAPHY

Aagaard, P. (2021). "Sediment Remineralisation in Temperate Estuaries". PhD thesis. University of Westhaven.

Blue Carbon Working Group (2023). *Guidelines for Blue Carbon Accounting in National Inventories*. Technical report BCWG-TR-07. Coastal Carbon Partnership.

Fairbairn, E. et al. (2022). "How permanent is blue carbon? A review of disturbance and remineralisation in vegetated coastal sediments". In: *Global Change Biology* 28.14, pp. 4301–4320. DOI: 10.0000/gcb.16215.

Holmgren, A. and K. Lund (2019). "Organic carbon burial in Scandinavian eelgrass sediments over the last two centuries". In: *Estuarine, Coastal and Shelf Science* 224, pp. 112–124. DOI: 10.0000/ecss.2019.04.017.

Lindqvist, M., T. Ferreira, and A. Holmgren (2024). "Open-top warming chambers for subtidal vegetation". In: *Limnology and Oceanography: Methods* 22.2, pp. 58–71. DOI: 10.0000/lom3.10590.

Marsh, R., C. Okafor, and T. Bell (2021). "Marine heatwaves in shallow temperate seas: frequency, duration and drivers". In: *Journal of Geophysical Research: Oceans* 126.7, e2020JC016942. DOI: 10.0000/2020JC016942.

Meer, J. van der and F. Achterberg (2016). "Segmented regression for detecting ecological thresholds". In: *Methods in Ecology and Evolution* 7.11, pp. 1302–1311. DOI: 10.0000/2041-210X.12611.

Okafor, C. and R. Marsh (2018). "Thermal tolerance of *Zostera marina* across a latitudinal gradient". In: *Marine Ecology Progress Series* 601, pp. 77–92. DOI: 10.0000/meps12671.

Rowan, H. (2017). *Seagrass Ecology: Structure, Function and Change*. 2nd ed. Northmoor: Northmoor University Press.

Sato, K. and E. Fairbairn (2023). "Recovery dynamics of eelgrass meadows after the 2018 North Sea heatwave". In: *Marine Biology* 170, p. 45. DOI: 10.0000/s00227-023-04190-2.

Tetley, I., S. Brannigan, and A. Holmgren (2020). "Dating recent seagrass sediments with excess lead-210: pitfalls and a protocol". In: *Limnology and Oceanography: Methods* 18.3, pp. 101–117. DOI: 10.0000/lom3.10352.